

B.Tech. Degree III Semester Regular/Supplementary Examination

January 2023

ME 19-205-0302 ELECTRICAL TECHNOLOGY

Time: 3 Hours

(2019 Scheme)

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Study the different types, constructional details, operational principles, and performance characteristics of DC motors and DC generators
- CO2: Understand the construction and working of transformers, transformer losses, current transformer, and potential transformers
- CO3: Understand the constructional details, operational principles, and performance characteristics of induction motors and alternators.
- CO4: Learn about the generation, transmission, and distribution of electrical energy.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) What is armature reaction in DC generator? How it affects the main flux distribution and how can armature reaction be reduced?	3	L1	1	1
(b) Which DC motor is preferred in hoists? Justify your answer.	3	L3	1	2
(c) Explain the statement with necessary equations "All power in autotransformer is not transformed inductively."	3	L2	2	1
(d) Draw the phasor diagram of a practical transformer which is connected to an inductive load.	3	L1	2	1
(e) Why synchronous motor is not self-starting? Suggest a starting method for the same.	3	L2	3	1
(f) A 2 pole, 3-phase induction motor runs at 2910 rpm on a 50 Hz supply. Find: (i) synchronous speed (ii) frequency of rotor emf.	3	L2	3	2
(g) Discuss corona effect in transmission lines.	3	L1	4	1
(h) Explain the function of economizer and super heater in a thermal power plant.	3	L2	4	1

PART B

(4 × 12 = 48)

II. (a) A 4 pole lap wound DC generator supplies to 50 lamps of 100 watts, 200 V each. The field and armature resistances are 50 Ω and 0.2 Ω respectively. Allowing a brush drop of 1V across each brush. Calculate: (i) Armature current (ii) Generated EMF (iii) Power output of DC armature.	4	L4	1	1
(b) Discuss the various methods of speed control of DC series motor with neat sketches	8	L2	1	1

OR

III. (a) During Swinburne's test, a 250 V DC machine was drawing 3 A from the 250 V supply. The resistances are 250 Ω and 0.2 Ω. Find the constant loss of the machine. Also find the efficiency of the machine when it is delivering a 20 A at 250 V.	8	L4	1	2
(b) How back emf is generated in DC motor? Explain its significance.	4	L2	1	1

(P.T.O.)

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		Marks	BL	CO	PO
		12	L4	2	1
IV.	The OC and SC test for a single phase, 5 kVA, 500/250 V, 50 Hz transformer gave the following results OC test: 500 V 1 A 50 W SC test: 25 V 10 A 60 (i) Determine efficiency on 60% full load 0.8pf leading (ii) At what load or kVA the transformer is to operated for maximum efficiency. (iii) Draw the equivalent circuit with respect to primary				
OR					
V. (a)	A 600 kVA single phase transformer working at unity power factor has an efficiency of 95 percent at both half full load and full load. Determine the efficiency at 70 percent of full load.	8	L4	2	1
(b)	Derive EMF equation of transformer and show that emf/turn is same for both primary and secondary windings.	4	L2	2	1
VI.	A 6 pole, 50 Hz three phase induction motor running on full load develops a useful torque of 160 Nm when the rotor emf makes 120 complete cycles per minute. The mechanical torque lost in friction is 5 Nm. Total stator loss is equal to 850 W. Calculate: (i) Rotor slip (ii) Rotor copper loss (iii) Shaft power output and (iv) Efficiency at full load.	12	L3	3	2
OR					
VII. (a)	Show that the maximum torque of an induction motor is not dependent on rotor resistance.	6	L3	3	2
(b)	A three phase, 6 poles, 60 Hz alternator having 200 conductors per phase and 10 mWb flux per pole. If the alternator winding is short circuited by 60°, then determine EMF induced per phase in the alternator. Assume distribution factor as 1.	6	L2	3	2
VIII. (a)	Discuss the working of a suitable power plant which requires less quantity of fuel for generation of given amount of energy compared to other conventional sources with a neat diagram. Also mention its advantages and disadvantages.	9	L4	4	1
(b)	Explain load dispatching.	3	L1	4	1
OR					
IX. (a)	List out the function of electrical equipment used in substations.	7	L2	4	1
(b)	Discuss the advantages of high voltage transmission. State its limitations too.	5	L1	4	1

Bloom's Taxonomy Levels

L1 = 14.28%, L2 = 34.45%, L3 = 17.6%, L4 = 34.45%.
